Product Development of a Substrate Mixer and Portioning Device

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Bachelor’s thesis
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Substrate for mushroom cultivation

Mix and water agricultural waste products

Bag, pasteurize and insert mushroom spawn

Harvest edible mushrooms

D MAVT GHE GLOBAL HEALTH ENGINEERING

Why should mushrooms be cultivated in developing countries?

Circular Economy: "From waste to money"

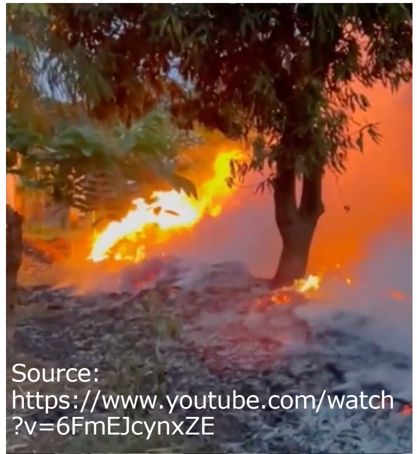
Agricultural waste

Waste disposal

Environmental pollution

Mushroom growing

Income generating



Source: https://www.youtube.com/watch?v=6FmEJcynxZE

How should mushrooms be cultivated in developing countries?

Ugandan baseline

Manual substrate preparation:

- High labor cost
- Inconsistent yields

Mechanized substrate preparation:

- High capital requirement
- High energy consumption

Industrial processing

How should mushrooms be cultivated in developing countries?

Ugandan baseline

Constraints in Kampala:

- Low capital (smallholders)
- Limited material availability
- Unreliable energy supply
- High energy cost

Goal of this project →

Industrial processing

Requirements for the device (Research Questions)

Mixing
How can heterogeneous agricultural residues be processed to achieve a homogeneous substrate composition throughout the mixing vessel?

Watering

How can the substrate be uniformly hydrated to achieve a target moisture content of 60-70 wt%?

Portioning

How can the processed substrate be portioned into bags with deviations within a tolerance of ±10 wt%?

Product Development - Methodology

Reproducibility

"Simplicity is the ultimate sophistication." – Leonardo da Vinci

Globally available materials

Simple manufacturing

Low-tech & low-cost

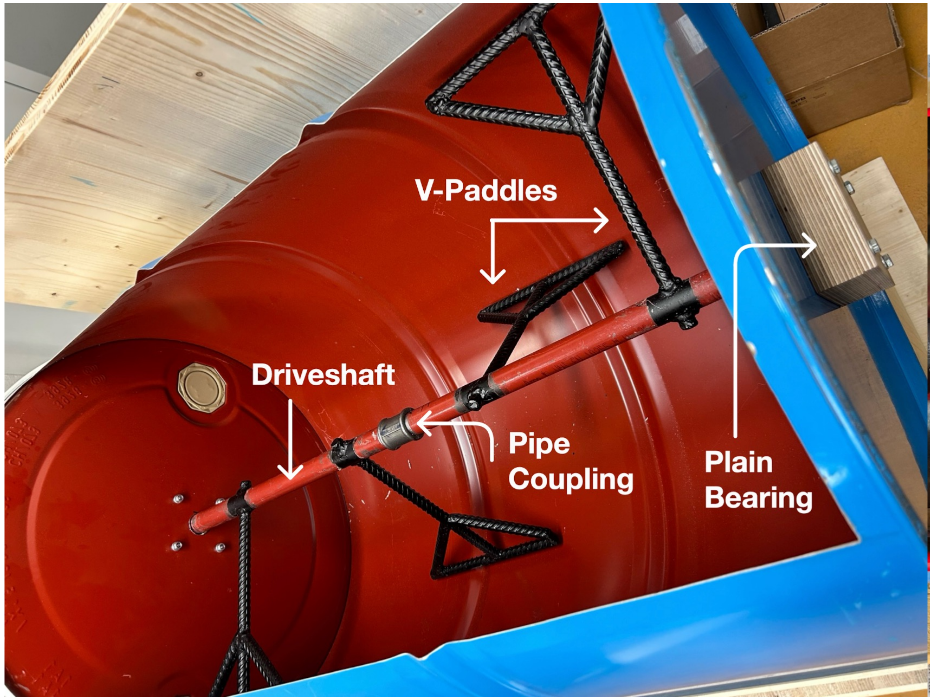
Product Development

- Oil drum
- Wooden panels

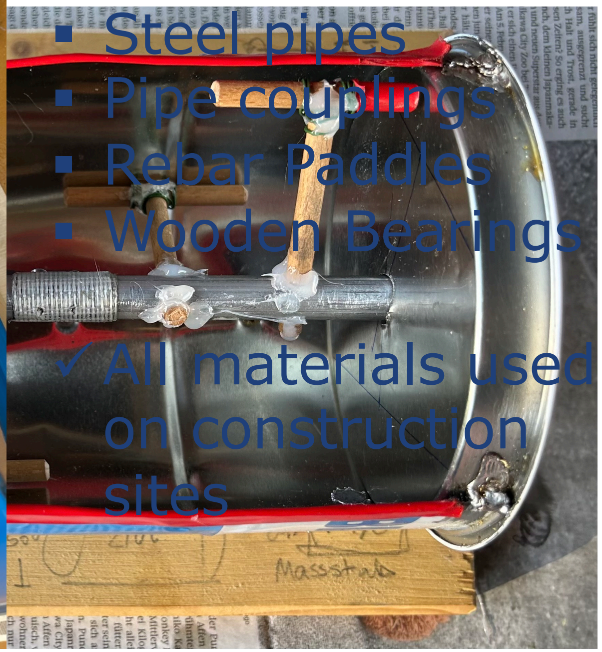
✓ Globally standardized materials & items

Product Development - Mixing

Final agitator



Model scale 1:5 – Testing different paddle geometries



- Steel pipes
- Pipe couplings
- Rebar Paddles
- Wooden Bearings

✓ All materials used on construction sites

Product Development - Watering

- Hydrostatic Reservoir
- Manifold (sprinkler)

✓ Gravity driven
(no energy supply)

Product Development - Portioning

Sealing Valve

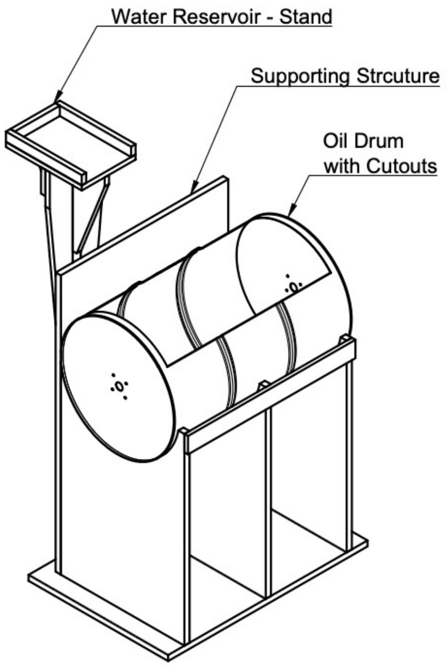
Conveyance Duct

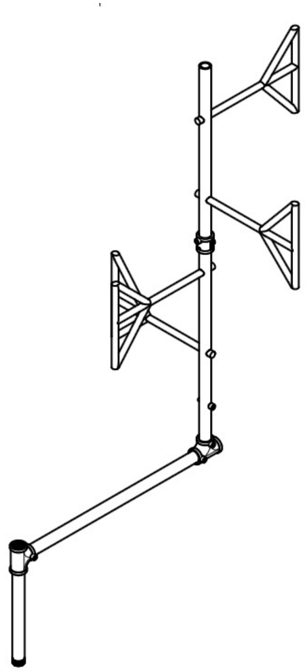
Slide Valve

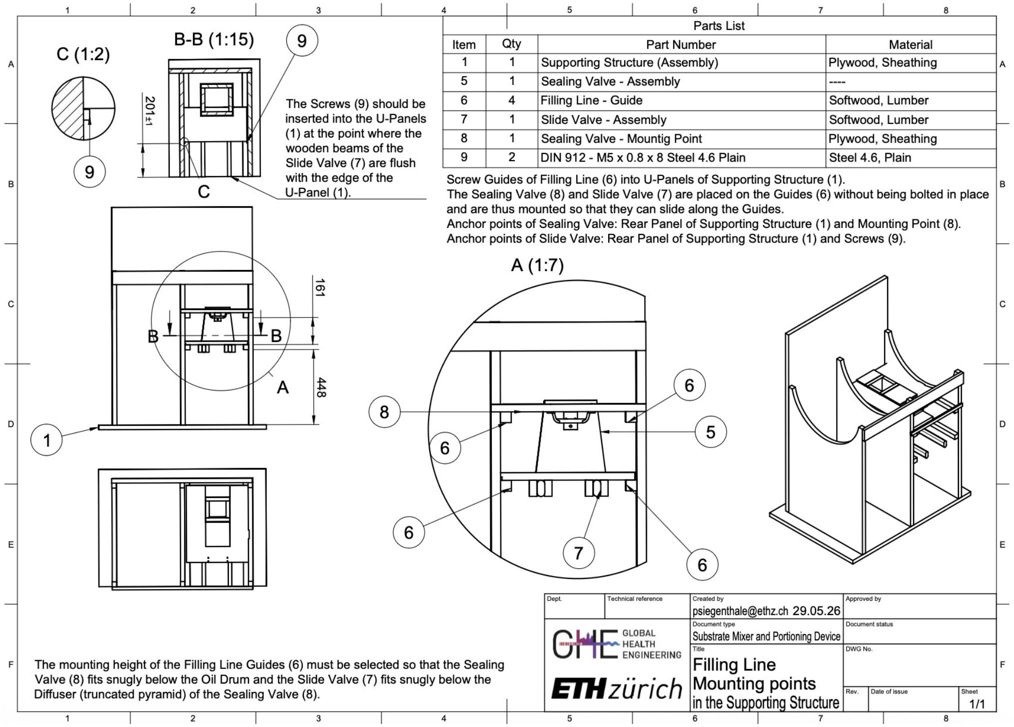
Product Development - Portioning

Substrate blockages in conveyance duct

Deliverables – Technical Drawings, User Manual

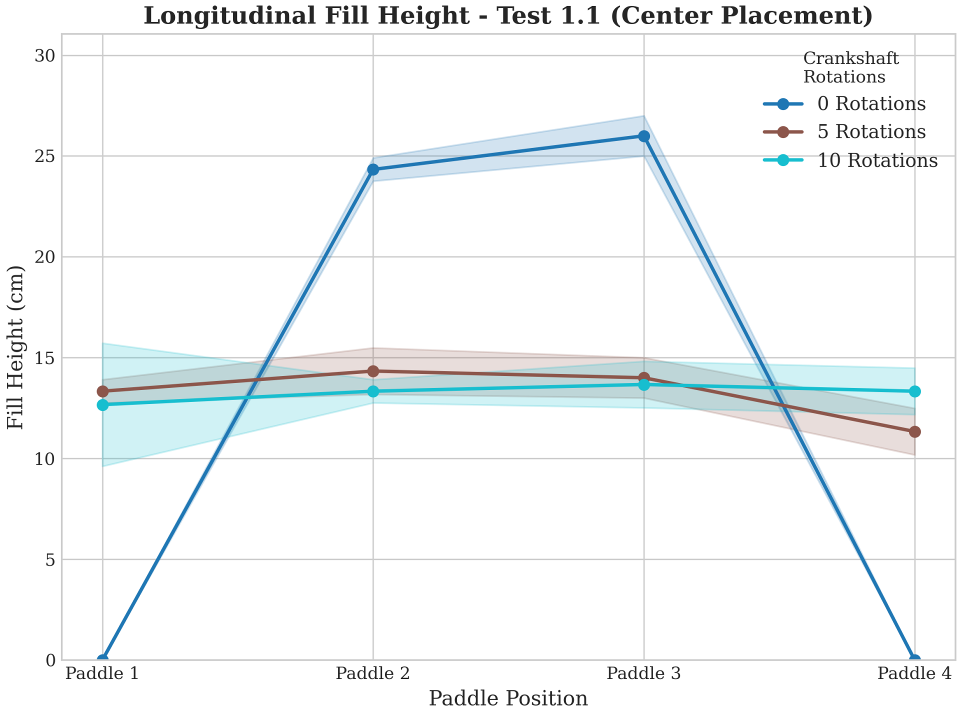


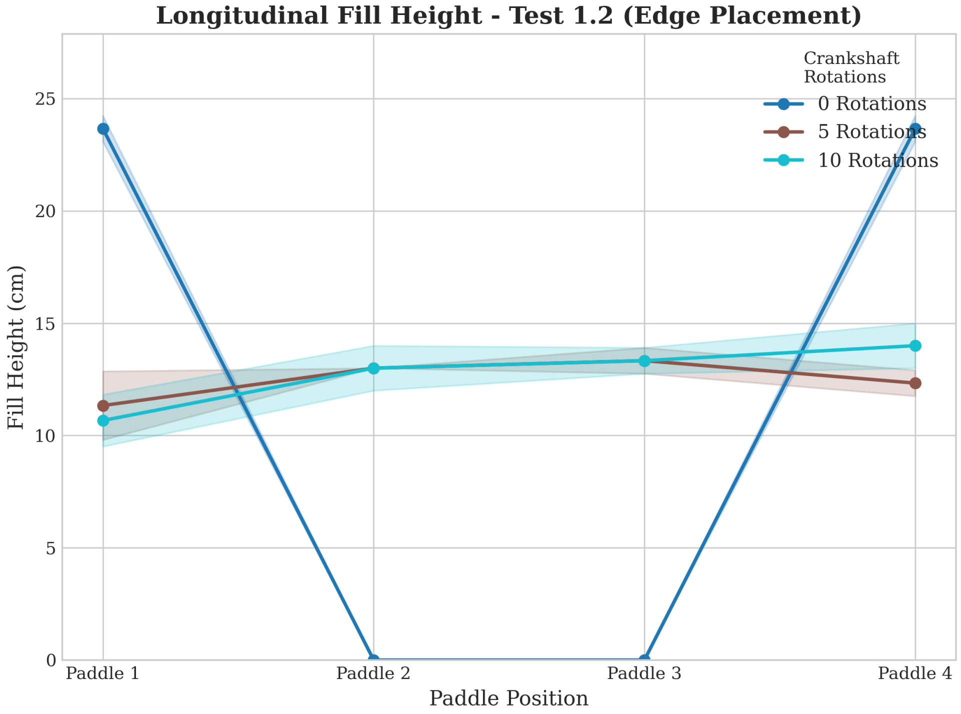




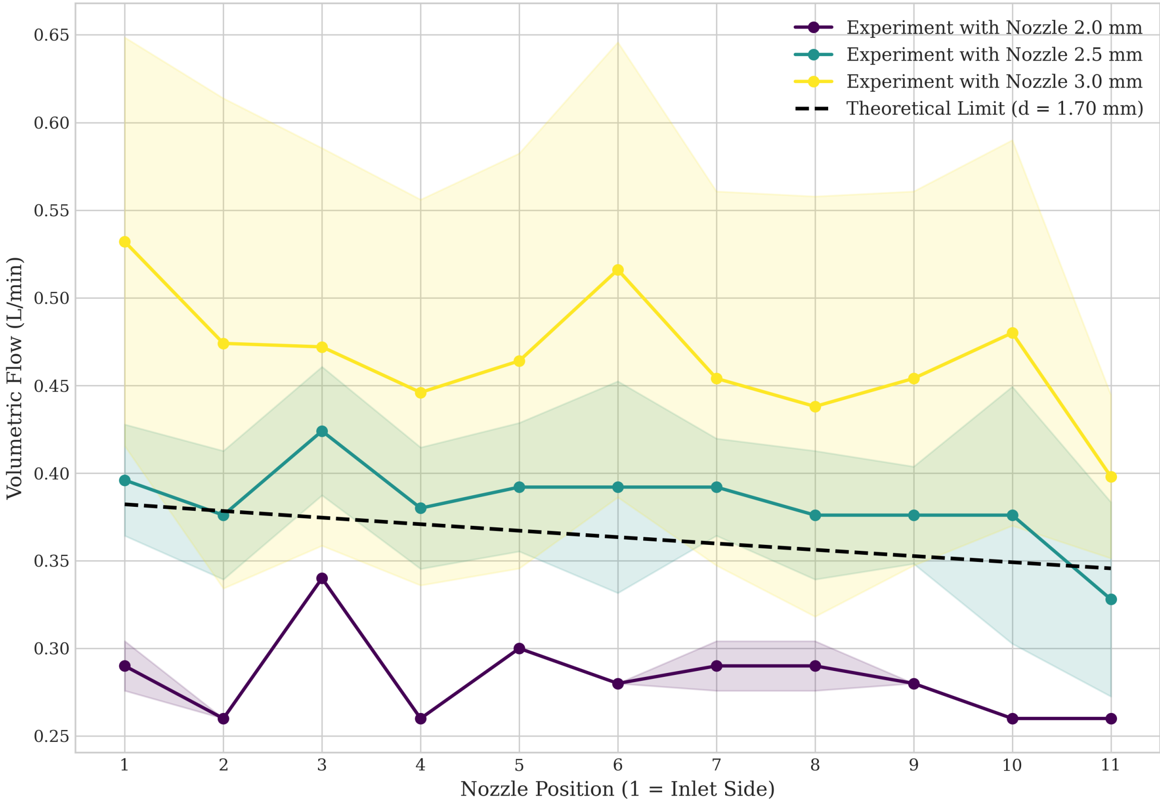
Item	Qty	Part Number	Material
1	1	Supporting Structure (Assembly)	Plywood, Sheathing
5	1	Sealing Valve - Assembly	----
6	4	Filling Line - Guide	Softwood, Lumber
7	1	Slide Valve - Assembly	Softwood, Lumber
8	1	Sealing Valve - Mountig Point	Plywood, Sheathing
9	2	DIN 912 - M5 x 0.8 x 8 Steel 4.6 Plain	Steel 4.6, Plain

Mixing – Longitudinal Transport

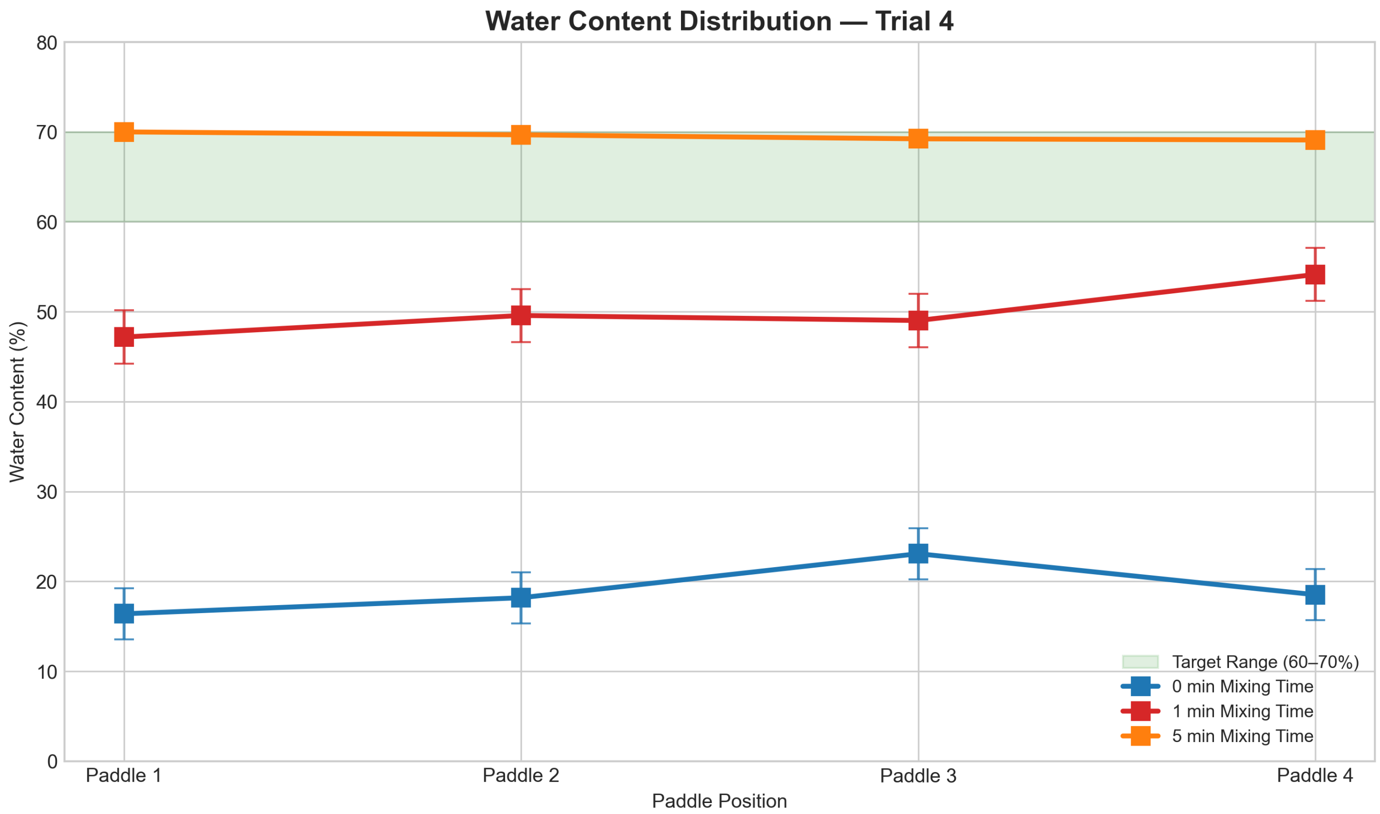




Watering – Nozzle Outflow



Mixing & Watering – Water Content Distribution



Mixing Vessel

(69.50 ± 0.41) wt%

Portioned Bags

(69.55 ± 0.62) wt%

Samples deviation (5’)

(0.45 ± 0.06) wt%

Portioning – Bagging Performance

Trial	Prototype	Mean (kg)	SD (kg)	RSD (%)	Loss (wt%)	Note
1	PE-Pipe	0.414	0.153	36.89	45.42	all bags
2	Diffuser	1.367	0.105	$\boxed{7.69}$	8.74	all bags
3	Diffuser	1.146	0.186	16.25	3.24	all bags
3	Diffuser	1.235	0.105	$\boxed{8.52}$	—-	bags 6–18

Conclusion – Working Prototype

Hand-driven, low-cost mechanical solutions can successfully standardize substrate processing.

Manual Baseline

Big reduction of labor estimated

Mechanized & Standardized Solution

Outlook – Recommendations

- **Field Validation in Uganda:**
➢ Analysis of user acceptance, reproducibility, durability.

ETH zürich

Final Prototype